Department of Electrical and Electronic Engineering
University of Peradeniya
Deep Vision. Bright Future

EEES

PeraBots '26

RULEBOOK
UNIVERSITY CATEGORY

DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING
UNIVERSITY OF PERADENIYA

1.INTRODUCTION

1.1 ABOUT PERABOTS 2026

PeraBots 2026 is the flagship annual robotics competition organized by the Electrical and Electronic Engineering Society (EEES), Department of Electrical and Electronic Engineering, University of Peradeniya, in collaboration with the Robotics Society of the University of Peradeniya. The competition aims to promote innovation, creativity, and practical engineering skills by providing participants with an opportunity to design, program, and develop autonomous robotic systems.

PeraBots 2026 features separate School and University categories, enabling participants of different experience levels to engage in robotics challenges appropriate to their academic background. Through a combination of simulation-based activities, practical implementation, and technical learning opportunities, the competition seeks to inspire the next generation of engineers, innovators, and technology enthusiasts while fostering collaboration among educational institutions across Sri Lanka.

1.2 UNIVERSITY CATEGORY OVERVIEW

The PeraBots 2026 University Category is designed to challenge undergraduate students to apply their knowledge in autonomous robotics through a series of practical engineering challenges. Participants will develop intelligent robotic systems capable of sensing, navigation, decision-making, and autonomous operation while strengthening their skills in system integration, programming, and problem solving. The competition provides a platform for students to translate theoretical knowledge into practical solutions within a competitive and collaborative environment.

1.3 COMPETITION FORMAT

The competition consists of two phases. **Phase 1** is conducted online using the Webots simulation platform, where teams will develop and program an autonomous virtual robot to complete a series of simulation-based challenges.

Teams will submit their simulation files for evaluation. After that the highest-performing teams will qualify for the final round.

In **Phase 2**, shortlisted teams will design and implement a physical robot to compete at the University of Peradeniya. The final round evaluates each team's ability to successfully transfer their simulation concepts into a real-world robotic system capable of completing the assigned challenges. Detailed task descriptions, technical specifications, and evaluation criteria for both phases are provided in the subsequent sections of this rulebook

1.4 IMPORTANT DATES

- Registration Closes
  September 22
- Phase 1 Begins (Opening Ceremony – Online)
  September 26
- Workshop Series (Online)
  September 28 – Octomber 12
- Phase 1 Submission Deadline
  October 28
- Finalists Announced
  November 5
- Phase 2 - Grand Finale & Closing Ceremony – Physical
  December 12

2.GENERAL COMPETITION INFORMATION

2.1 ELIGIBILITY

- The University Category is open to **undergraduate students** from recognized universities and higher education institutes in Sri Lanka.
- All members of a team must represent the **same university or higher education institute**.
- Each participant may represent **only one team** throughout the competition.

2.2 TEAM COMPOSITION

- A team may consist of **1 to 5 members**.
- All team members must register under a single university or higher education institute.
- A university or higher education institute may register **multiple teams** under its name.
- Each team shall nominate **one Team Leader**, who will serve as the primary contact for all competition-related communication.
- Each team may submit **only one robot** for the competition.

2.3 REGISTRATION PROCEDURE

- Teams must register through the **official Google Registration Form** before the registration deadline.
- All required information must be completed accurately during the registration process.
- The **registration number** of every participant is mandatory during registration.
- Teams must upload **a single PDF containing scanned copies of the valid University Student Identity Cards of all team members** during registration.
- Once the registration period has ended, **changes to team members will not be permitted** under any circumstances.
- Participant names submitted during registration will be used for all official documentation, including certificates. Teams are responsible for ensuring that all information provided is accurate.

The organizing committee reserves the right to verify the eligibility of any participant or team at any stage of the competition. Failure to provide valid supporting documents upon request may result in disqualification.

2.4 OFFICIAL COMMUNICATION

- Upon successful registration, the Team Leader will receive links to the official **PeraBots 2026 WhatsApp Community and WhatsApp Announcement Channel**.
- The Team Leader is required to join both platforms and ensure that all team members are added to the WhatsApp Community or provided with the invitation link.
- **All official announcements, updates, notices, schedule changes, and competition-related communications will be disseminated through these official WhatsApp platforms.**
- Teams are responsible for regularly checking these platforms and remaining informed of all announcements issued by the organizing committee.

2.5 CODE OF CONDUCT

- Participants are expected to maintain professionalism, honesty, and respectful behavior throughout the competition.
- Teams shall comply with all competition rules, deadlines, and instructions issued by the organizing committee.
- Any attempt to gain an unfair advantage, provide false information, or engage in misconduct may result in disqualification.
- The decisions of the organizing committee and judging panel shall be considered final in all competition matters.

3.COMPETITION FORMAT

3.1 COMPETITION FLOW

The University Category of PeraBots 2026 consists of two consecutive phases: an **Online Simulation Round** and a **Physical Final Round**. All registered teams are required to participate in the Online Simulation Round. Teams demonstrating the highest overall performance in this phase will qualify for the Physical Final Round, where they will implement and compete using a physical robot in a real-world competition arena.

3.2 PHASE 1 – ONLINE SIMULATION ROUND

The Online Simulation Round will be conducted using the **Webots** robotics simulation platform. Teams will complete the assigned competition tasks by designing and programming an autonomous virtual robot within the provided simulation environment. Upon completion, teams must submit the required simulation files through the official submission platform before the specified deadline. Further details regarding the competition tasks, submission requirements, and evaluation criteria are provided in the subsequent sections of this rulebook

3.3 PHASE 2 – PHYSICAL FINAL ROUND

Teams selected from the Online Simulation Round will be invited to participate in the Physical Final Round at the University of Peradeniya. Finalists will design, build, and program a physical autonomous robot capable of completing the assigned competition tasks in a specially prepared competition arena. Additional rules, technical specifications, and competition procedures applicable to the Final Round are provided in the relevant sections of this rulebook.

3.4 QUALIFICATION FOR THE FINAL ROUND

Following the evaluation of all Phase 1 submissions, **a selected number of teams demonstrating the highest overall performance** will qualify for the Physical Final Round. The list of qualified teams will be officially announced **within one week after the Phase 1 submission deadline** through the official PeraBots 2026 communication platforms.

The organizing committee reserves the right to make the final decision regarding team qualification. The decisions of the judging panel and organizing committee shall be final and binding.

4. PHASE 1 – ONLINE SIMULATION ROUND

4.1 WEBOTS PLATFORM AND COMPETITION PACKAGE

Phase 1 of the University Category will be conducted using the **Webots robotics simulation platform**. The official competition package will be provided through the download link included in this rulebook. Participants must download and extract the provided package and open the designated Webots world file using a compatible version of Webots.

The package will contain the competition arena, the base robot model, and the necessary simulation environment. Teams are responsible for configuring the robot, integrating the required sensors and indicators, and developing the controller necessary to complete the assigned tasks.

Competition Package:Download from here
Required Webots Version: [R2025a] Cyberbotics: Robotics simulation with Webots

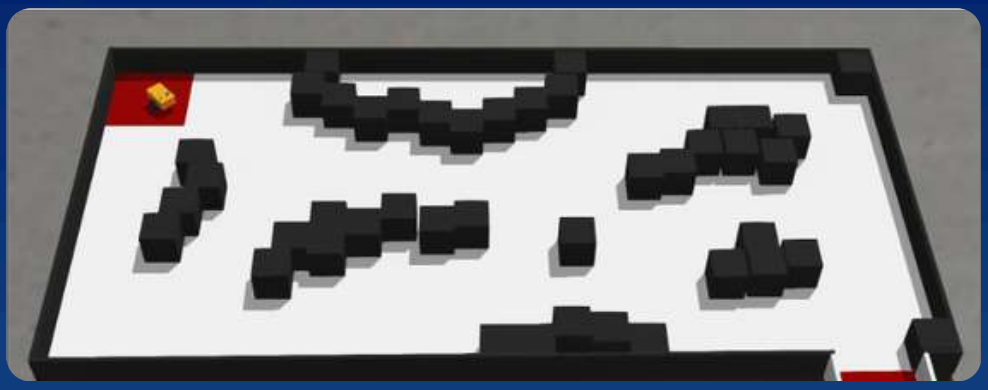
4.2 COMPETITION TASKS

The Phase 1 challenge consists of two consecutive sections, designed to evaluate autonomous navigation, environmental sensing, maze exploration, and decision-making.

SECTION 1 – AUTONOMOUS OBSTACLE NAVIGATION

The robot will begin inside a **25 cm × 25 cm starting zone** and must autonomously navigate through an obstacle field to reach the transition zone leading to Section 2.

The obstacle field will be constructed using fixed **10 cm × 10 cm × 10 cm cubes**. Multiple cubes may be arranged together to form larger and irregularly shaped obstacles. One or more feasible routes will be available through the arena, with a **minimum clear passage width of 25 cm** along intended traversable paths.



The robot must identify and navigate a suitable route without manual intervention while minimizing travel distance, completion time, and collisions with obstacles.

SECTION 2 – AUTONOMOUS MAZE EXPLORATION AND DECISION-MAKING

Upon completing Section 1, the robot will enter a **4 × 9 cell maze consisting of 36 cells**.

Each maze cell will have a **cell pitch of 25 cm × 25 cm**, measured from wall centerline to wall centerline, with maze walls approximately **1.2 cm thick**. Therefore, the clear internal area available within each cell will be approximately **23.8 cm × 23.8 cm**.

Blue and green colored tiles will be located at selected dead ends within the maze. Not every dead end will necessarily contain a colored tile, and the total number and distribution of colored tiles will not be disclosed in advance.

The robot must autonomously explore the maze sufficiently to identify the colored tiles, determine the number of blue and green tiles, and establish which color represents the **majority**. Efficient exploration is encouraged, and unnecessary repetition of previously explored paths will contribute to the total distance traveled.

After completing its exploration, the robot must navigate to the designated end cell. The robot will be considered to have reached the end cell only when its **entire body is positioned within the clear internal area of the designated cell**.

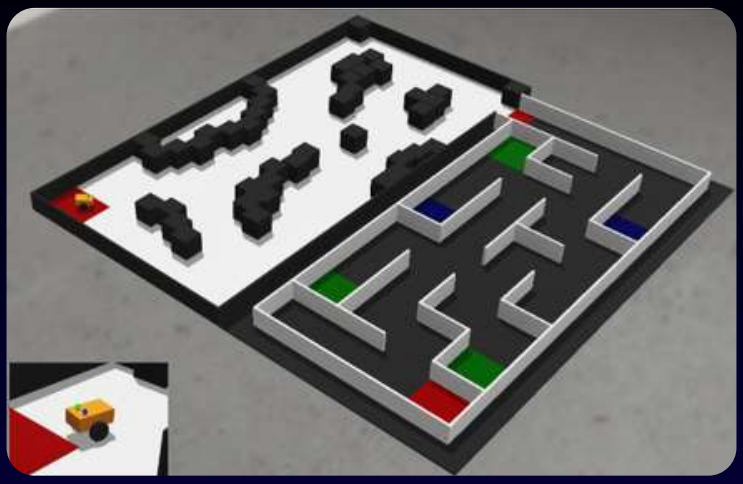
Within **3 seconds** of reaching the end cell, the robot must illuminate the LED corresponding to the majority color and come to a stop.



4.3 ARENA AND SIMULATION ENVIRONMENT

The Phase 1 arena consists of the obstacle-navigation area described in Section 1 and the maze environment described in Section 2.

In the Webots package provided to participants, the arena structure, including the floors, walls, boundary markers, and maze tiles, is packaged as a single locked node in the scene tree and labeled Arena. The Section 1 obstacles are similarly packaged as a separate locked node labeled Obstacles.



These nodes are locked intentionally to prevent accidental modification of the competition environment during development and testing. Participants must not unlock, edit, replace, or otherwise modify these nodes, including their internal geometry, wall positions, or obstacle placements, in any submitted project.

The following dimensional specifications will remain the same in both the development package and the official evaluation environment:

- **Section 1 floor area:** 2.44 m × 1.22 m, including a 25 cm × 25 cm starting zone and a marked transition zone leading to Section 2.
- **Obstacles:** Fixed 10 cm × 10 cm × 10 cm cubes, with a minimum clear passage width of 25 cm maintained along all intended traversable paths.
- **Section 2 maze:** 9 × 4 grid consisting of 36 cells, with a 25 cm × 25 cm cell pitch measured from wall centreline to wall centreline. Maze walls will be approximately 1.2 cm thick, resulting in a clear internal area of approximately 23.8 cm × 23.8 cm per cell.

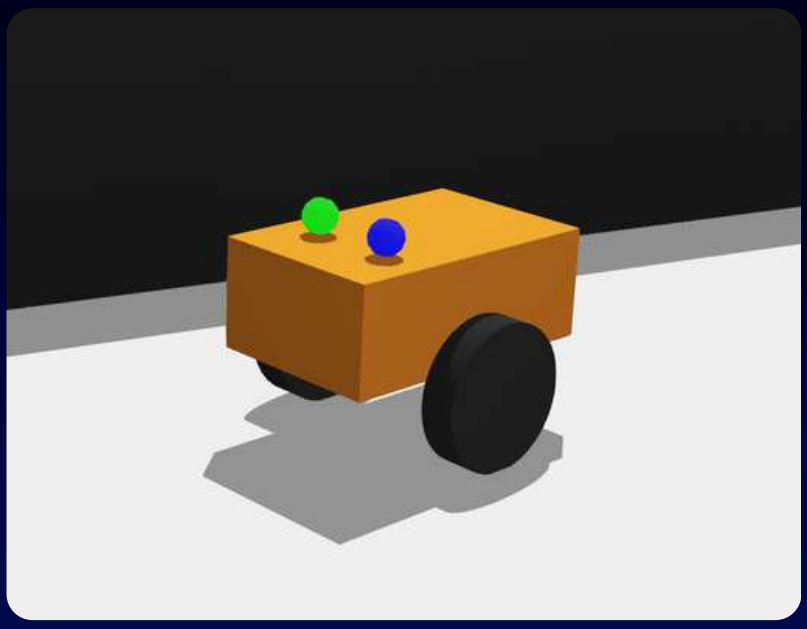
The simulation environment supplied to participants is intended for development, testing, and familiarization. The official evaluation environment will use the same locked **Arena** and **Obstacles** structure and the same dimensional specifications stated above.

However, the official evaluation environment will contain a different obstacle arrangement in Section 1 and a different internal wall and coloured-tile configuration in Section 2. All fundamental arena conditions specified in this rulebook will remain unchanged.

The same official evaluation configuration will be used for all participating teams to ensure fairness.

Teams are therefore expected to develop generalized autonomous solutions and must not rely on a particular obstacle arrangement, maze configuration, coloured-tile location, or predetermined route

4.4 ROBOT CONFIGURATION AND TECHNICAL REQUIREMENTS



- The robot must remain within a maximum overall dimension of **20 cm × 20 cm × 20 cm.**
- Teams must use the **base robot PROTO** provided within the official Webots competition package.
- The base robot's chassis, motors, wheel dimensions, physics properties, collision geometry, and other mechanical parameters must remain unchanged.
- Teams may make **minor positional adjustments to the two main drive wheels only,** where necessary to maintain proper robot balance after sensor placement. Such adjustments must not alter the wheel size, motor characteristics, or any other physical properties of the robot.
- The base robot is provided with two pre-installed LED devices named led_blue and led_green for the final colour decision in Section 2.
- The required LEDs must not be removed, renamed, repositioned for another purpose, or repurposed for any function other than the intended majority-colour indication.
- The robot controller must activate the appropriate LED, led_blue or led_green, once the majority colour has been determined.

Teams may configure the robot and add permitted sensors or devices through the designated robot extension point, subject to the following requirements:

- **Permitted devices:** DistanceSensor, Camera, TouchSensor, Accelerometer, Gyroscope, InertialUnit, LightSensor, PositionSensor for any added joints, Speaker, and additional LED devices for debugging or cosmetic purposes.
- **Prohibited devices:** GPS, Emitter, and Receiver.
- Supervisor functionality is disabled for the base robot and must not be enabled.

Teams may modify permitted robot configurations, including sensor placement and orientation, chassis colour, and controller logic, provided that all locked components and the required LEDs remain unchanged.

Once an evaluation attempt begins, the robot must operate fully autonomously without manual control, external commands, or participant intervention.

Solutions must not rely on hard-coded knowledge of the official evaluation arena, including obstacle positions, maze-wall configuration, coloured-tile locations, or predetermined routes.

The organizing committee reserves the right to inspect submitted project files, robot configurations, and controller code to verify compliance with the competition rules, including confirmation that the locked base-robot components and required LEDs have not been altered.

4.5 SUBMISSION REQUIREMENTS

Each team must submit its completed **Webots project** through the official submission method specified by the organizing committee before the submission deadline.

The submitted project must contain all files required to open, inspect, and execute the team's solution, including:

- The complete project folder used for development and testing, including the unmodified Arena.proto and Obstacles.proto files, the team's modified TwoWheelRobot.proto, and the complete controllers/main_controller/ folder.

- Any additional files required by the team's implementation, such as helper modules, configuration files, or trained models. These files must be placed inside the controllers/main_controller/ folder so that they are included during official evaluation.

The following file and folder names must be preserved exactly as provided in the original competition package:

- TwoWheelRobot.proto
- controllers/main_controller/
- main_controller.py

Teams may modify the **contents** of the permitted files within the customization limits defined in Section 4.4. However, these files and folders must not be renamed, moved, or restructured.

During official evaluation, the evaluation system will use the submitted TwoWheelRobot.proto together with the complete controllers/main_controller/ folder in the official evaluation environment. Submissions that do not follow the required file names or folder structure may fail to load and will not be manually corrected by the evaluation team.

Since the required internal file names are identical for all teams, each submission must be clearly identified using the t**eam name or team ID at the outer submission level.**

For example:

TeamName_Submission.zip

Teams must not add their team name or team ID to the required internal file names.

The submitted project must run correctly within the specified Webots environment without requiring modification by the evaluation team.

Detailed submission instructions, final naming conventions, the required archive format, and the official submission link will be communicated through the PeraBots 2026 official communication platforms.

4.6 EVALUATION CRITERIA

All submitted projects will be evaluated by the PeraBots 2026 evaluation team using the **official evaluation environment**. Participants will not control the robot during this process.

SECTION 1: EVALUATION

Each robot will be provided with **three evaluation attempts** for Section 1. Performance will be assessed based on:

- Successful navigation from the starting zone to the Section 2 transition;
- Time taken to complete Section 1;
- Total distance travelled and resulting path efficiency;
- Number of collisions with obstacles; and
- Overall successful completion of the section

Any physical contact between the robot and a fixed obstacle will be recorded as a collision.

SECTION 2 : EVALUATION

Section 2 will also be evaluated through **three dedicated attempts.** If necessary, the evaluators will manually position the robot at the designated entrance to Section 2 so that its maze-exploration performance can be assessed independently of its Section 1 performance.

Evaluation will consider:

- Successful and sufficient exploration of the maze;
- Total time taken;
- Total distance travelled and exploration efficiency;
- Number of collisions with maze walls;
- Correct detection and interpretation of coloured tiles;
- Correct identification of the majority colour;
- Successful navigation to the designated end cell; and
- Correct LED indication within the specified time.

If the robot successfully transitions from Section 1 into Section 2 during a continuous run, this will **not replace or reduce the three dedicated Section 2 attempts**. Successful autonomous transition between the two sections will contribute additional marks.

Additional marks will also be awarded for a robot that successfully completes the **entire challenge continuously from the Section 1 starting zone to successful completion of Section 2 without evaluator intervention.**

Performance across all three attempts will contribute to the evaluation with defined weightings. The **best individual performance,** consistency across attempts, section-wise performance, and overall challenge performance will be considered in determining the final Phase 1 score.

Attempt Termination

An evaluation attempt may be terminated under either of the following conditions:

- The robot becomes stuck, repeatedly performs an ineffective action, enters a continuous loop, or otherwise makes **no meaningful progress for 15 consecutive seconds**; or
- The robot's **entire body remains completely outside the defined track area for more than 2 consecutive seconds.**

A terminated attempt will be recorded as an unsuccessful attempt

4.7 TIE-BREAKING CRITERIA

In the event that two or more teams achieve the same final Phase 1 score, ranking will be determined using the following criteria in order:

1. Successful completion of the entire challenge;
2. Higher overall task-completion performance;
3. Lower overall completion time;
4. Shorter total distance travelled;
5. Fewer recorded collisions.

If a tie remains after applying the above criteria, the final decision will be made by the PeraBots 2026 evaluation panel.

5. PHASE 2 – PHYSICAL FINAL ROUND

5.1 COMPETITION VENUE

The Physical Final Round of the PeraBots 2026 University Category will be conducted at the **University of Peradeniya.**

Teams selected based on their performance in Phase 1 will be invited to participate in the Final Round with their fully developed physical robots. Details regarding the competition schedule, reporting time, venue arrangements, and other logistical information will be communicated to qualified teams through the official PeraBots 2026 communication platforms.

5.2 COMPETITION ARENA

The Final Round arena will consist of **three main sections.**

The first two sections will be based on concepts similar to those introduced during the Phase 1 simulation challenge. These may involve autonomous navigation, obstacle avoidance, maze exploration, environmental sensing, and decision-making.

An **additional third section** will be introduced in the Final Round. Information detected, collected, or determined by the robot while completing the first and second sections may be required to make decisions or perform specific actions in the third section.

The exact arena layout, task sequence, environmental conditions, and requirements of the third section will not be disclosed in advance. Teams are therefore encouraged to develop adaptable robotic systems capable of sensing their environment, retaining and processing information obtained during earlier sections, and using that information for subsequent autonomous decisions.

Detailed specifications of the Final Round arena will be provided in the **Final Round Challenge Guide.**

5.3 ROBOT SPECIFICATIONS

The physical robot used in the Final Round must satisfy the following general requirements:

- The maximum overall dimensions of the robot shall be 20 cm × 20 cm × 20 cm.
- The robot must be capable of operating fully autonomously during an official competition run.
- The robot should be capable of performing the sensing, navigation, exploration, and decision-making functions introduced during Phase 1.
- Teams may select suitable sensors, actuators, controllers, processing units, and other electronic or mechanical components required for their implementation, subject to the competition rules.
- The robot must not intentionally damage, alter, mark, or displace any part of the competition arena.
- The robot must not contain mechanisms, components, or materials that may create an unreasonable safety risk to participants, officials, competition equipment, or the surrounding environment.
- Wireless or external control of the robot during an official run will not be permitted unless specifically allowed by the Final Round rules.
- The robot must comply with any additional technical and safety requirements specified in the Final Round Challenge Guide.

The organizing committee reserves the right to inspect any robot before an official run to verify compliance with the competition requirements.

5.4 COMPETITION PROCEDURE

Qualified teams must bring their completed physical robots to the University of Peradeniya and participate according to the schedule and instructions provided by the organizing committee.

The Final Round will extend the concepts introduced during Phase 1 into a physical competition environment. Robots will be required to progress through the arena autonomously and complete the assigned tasks without participant intervention.

During the challenge, the robot may be required to **collect, retain, and process information obtained in earlier sections and use that information to determine the appropriate action in a subsequent section.**

Once an official run has started, participants shall not manually control, touch, reposition, or otherwise assist the robot unless instructed or permitted by a competition official.

The detailed competition procedure, including the complete challenge description, arena configuration, task sequence, starting and completion conditions, number of attempts, and other operational requirements, will be specified in the **Final Round Challenge Guide.**

5.5 EVALUATION CRITERIA

Performance in the Final Round will be evaluated based on the robot's ability to autonomously and successfully complete the assigned challenge.

Evaluation may consider the following factors:

- Successful completion of individual sections and the overall challenge;
- Autonomous navigation and task execution;
- Correct sensing and interpretation of the competition environment;
- Ability to collect and utilize information obtained during different sections of the arena;
- Accuracy of autonomous decision-making;
- Completion time and navigation efficiency;
- Number of collisions or navigation errors; and
- Successful execution of the required final actions.

The detailed scoring system, marking distribution, and evaluation procedure will be provided in the **Final Round Challenge Guide.**

5.6 PENALTIES

Penalties may be applied for violations of the competition rules or specified operating procedures. These may include:

- Unauthorized manual intervention during an official run;
- Collisions with or improper interaction with arena elements;
- Failure to follow the specified starting or competition procedure;
- Violation of robot dimensional or technical requirements;
- Intentional modification, displacement, or damage to the competition arena;
- Use of prohibited external control or communication; and
- Any action that provides an unfair advantage over other teams.

Serious safety violations, intentional damage to competition equipment, or deliberate violations of the competition rules may result in termination of an attempt or disqualification.

The detailed penalty structure applicable to the Final Round will be provided in the **Final Round Challenge Guide.**

5.7 TIE-BREAKING PROCEDURE

In the event of a tie between two or more finalist teams, the final ranking will be determined according to the tie-breaking criteria specified for the Final Round.

The tie-breaking procedure may consider factors such as overall task completion, completion time, navigation efficiency, number of collisions, and performance in individual sections of the challenge.

The complete tie-breaking procedure will be specified in the **Final Round Challenge Guide**. Decisions made by the official judging panel regarding the Final Round results shall be final.

Final Round Challenge Guide

The complete Final Round challenge will **not be disclosed in this rulebook**. A separate **Final Round Challenge Guide** containing the detailed challenge description, arena configuration, task sequence, third-section requirements, scoring scheme, penalties, and specific competition procedures will be revealed to qualified teams during the **Opening Ceremony of the Final Round**.

Teams should therefore develop their physical robots based on the technical concepts introduced during Phase 1 while ensuring sufficient adaptability to respond to additional tasks and information introduced during the Final Round.

6. AWARDS & RECOGNITION

6.1 AWARDS

The top-performing teams in the PeraBots 2026 University Category will be recognized based on their Final Round performance.

The following awards will be presented:

- **Champions**
- **1st Runner-up**
- **2nd Runner-up**

The winning teams will receive cash prizes and medals in recognition of their achievement.

Details regarding the prize amounts and any additional awards will be announced through the official PeraBots 2026 communication platforms.

6.2 CERTIFICATES

All teams that qualify for and participate in the **Physical Final Round** will receive certificates in recognition of their participation and achievement in PeraBots 2026.

Certificates may also be issued to award-winning teams according to their final placement.

Further information regarding certificate distribution will be communicated through the official competition channels.

7. RULES & REGULATIONS

7.1 GENERAL RULES

- All participants must comply with the instructions provided in this rulebook and by the organizing committee.
- Robots must operate autonomously during official evaluation and competition runs unless otherwise stated.
- Participants must not interfere with the arena, another team, or the evaluation process.
- Teams are responsible for ensuring that their submitted files, software, and physical robots are functional and comply with the specified requirements.
- Any attempt to gain an unfair advantage, manipulate the evaluation environment, or violate the intended competition procedure may result in penalties or disqualification.
- Participants must follow all schedules, deadlines, and instructions communicated through the official PeraBots 2026 platforms.

7.2 SAFETY REQUIREMENTS

All teams must ensure that their robots and associated equipment are safe to operate.

Robots must not contain components, mechanisms, or materials that may cause injury, electrical hazards, fire risks, or damage to the competition arena or surrounding equipment.

The organizing committee reserves the right to inspect any robot and prevent it from participating if it is considered unsafe.

7.3 DISQUALIFICATION CONDITIONS

A team may be disqualified for serious violations including:

- Intentional damage to the competition arena or equipment;
- Use of prohibited manual or external control during an official run;
- Submission of false participant information or violation of eligibility requirements;
- Deliberate manipulation of the evaluation process;
- Repeated failure to follow instructions given by competition officials; or
- Any form of misconduct that significantly affects the fairness or safety of the competition.

The organizing committee may issue warnings or penalties for minor violations before considering disqualification.

7.4 JUDGES' DECISIONS

All evaluations, penalties, rankings, and competition results will be determined by the officially appointed evaluation and judging panels.

The decisions of the judges and the organizing committee regarding competition results, rule interpretation, and disputes shall be **final**.

9. UNIVERSITY CATEGORY COORDINATORS

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